

STREAMIFY

A Modern Peer-to-Peer Skill-Sharing & Language Exchange Social Network

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<https://github.com/mohamedahmedzaghlol/Streamify-Language-Exchange.git>

Project Overview & Core Mission

Providing bilateral skill matching and real-time structured social
education nodes

Problem Space vs. The Solution

The Core Problem

Contemporary social platforms suffer from high-volume distraction rather than structured education loops. Users face discovery mismatch, unstructured indexing layouts, and massive database overhead from nested relationship constraints.

The Streamify Solution

A decoupled MERN platform featuring strict relational modeling. Offloads real-time socket workloads to cryptographically secure infrastructure, bringing scalability, automated matches, and reliable performance.

Core Capability Pipeline



Smart Onboarding

Captures dynamic user skill sets, preferred language parameters, and geolocation vectors to set profile initialization parameters safely.



Discovery Engine

Enables filtered peer search based on inverse complementary skill desires (e.g., matching fluent Arab coders with English native learners).



Stateful Relations

Optimized database connection pools processing asynchronous request cycles (Pending, Incoming, Outgoing) cleanly.

Architectural Layout & Stack

- ✓ **Frontend UI Layer:** Designed via React and optimized using TanStack Query for reactive client-side state caching.
- ✓ **Backend Application Layer:** Node.js and Express.js REST APIs structuring modular clean-code pattern routes.
- ✓ **Database Collection:** Schema constraints mapped strictly on MongoDB with optimized object relationships.
- ✓ **Offloaded Engine:** Seamless Stream.io SDK handling messaging flows securely via programmatic credentials.



Technical Implementation Details

Deep-dive into Mongoose collection routing, API dependencies, and security setups

Secure & Optimized Mechanics



Isolated Schema References

Instead of nesting unbounded arrays inside User documents, relation items are recorded inside a dedicated lightweight collection, ensuring query operations stay consistent under load.



Cryptographic Token Signatures

The server signs real-time JWT communication strings using Stream's official backend SDK, preserving maximum endpoint security during direct client connections.



Safe Credential Execution

Encapsulating application secrets inside environmental files using dotenv, and encrypting database authentication structures using industry standard bcryptjs salting.

Overcoming Engineering Obstacles

1. Document Size Limits

The Obstacle: Growing friendship arrays within MongoDB users threatened the 16MB document boundary.

The Fix: Designed separate FriendRequest collections with distinct indexing schemas to ensure infinite horizontal growth.

2. DevOps Path Mapping

The Obstacle: Encountering module loader failures within multi-stage Docker configurations.

The Fix: Rebuilt customized environment mappings and isolated production package builds directly in the root context.

Operational REST API Layer

HTTP Method	Endpoint Target Path	Action Performed	Database Scope
POST	/api/auth/register	Saves credential hashes into User Schema	users (Insert)
POST	/api/auth/onboarding	Sets isOnboarded to true, maps skills	users (Update)
POST	/api/user/friend-request	Initializes pending relationship record	friendrequests (Create)
GET	/api/chat/token	Generates cryptographic JWT communication keys	None (Algorithmic)

Preliminary Progress Overview

Backend Core & Schema

100%

REST API Route Logic

100%

Postman Validation Tests

100%

Client Integration (React)

40%

Development tracking chart indicating fully operational backend APIs, database persistence models, and completed SDK tests.

Development Roadmap

Milestone 1

Database modeling, schema indexing constraints, setup validation loops.

Milestone 2

Building secure Express controllers & Stream cryptographic endpoints.

Milestone 3

Full-scale integration tests over Postman, bug fixes, GitHub push.

Milestone 4

Frontend bindings integration, containerization, cloud deploy.

Questions & Answers

Thank you for your time and guidance!

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Image Sources



https://static.vecteezy.com/system/resources/previews/046/567/608/non_2x/abstract-peacock-blue-ocean-dark-blue-technological-modern-flowing-wave-line-background-teal-blue-and-black-digital-future-technology-web-banner-background-for-presentation-cover-slide-magazine-vector.jpg

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